

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 2

d_S Y_S d_F Y_F N	η	H η	N D
$2 \quad \frac{1}{2} \quad 1 \quad 0 \quad -\frac{1}{2} M_N N_{r_1}{}^T C N_{r_1}$	$+\frac{1}{2} \lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}{}^\dagger$	$+\frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+\frac{1}{2} \lambda_5 \eta_{a_1} \eta_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$ $+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$	$+\frac{1}{2} i N_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
$2 \quad \frac{1}{2} \quad 3 \quad 0 \quad -\frac{1}{2} M_N N_{A_1, r_1}{}^T C N_{A_1, r_1}$	$+\frac{1}{2} \lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}{}^\dagger$	$+\frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+\frac{1}{2} \lambda_5 \eta_{a_1} \eta_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$ $+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}{}^\dagger \eta_{a_2}{}^\dagger$	$+\frac{1}{2} i N_{A_1, r_1}{}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1, r_1})$
$3 \quad \frac{1}{2} \quad 2 \quad 0 \quad -M_N \overline{N_{a_1, r_1}} N_{a_1, r_1}$	$+\frac{1}{2} \lambda_S \eta_{A_1} \eta_{A_2} \eta_{A_1}{}^\dagger \eta_{A_2}{}^\dagger$ $+M_\eta^2 (-\eta_{A_1}) \eta_{A_1}{}^\dagger$	$+\lambda_3 H_{a_1} \eta_{A_1} H_{a_1}{}^\dagger \eta_{A_1}{}^\dagger$	$+i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$

Field combinations 2 of 2

d_S	Y_S	d_F	Y_F	ηD	$\eta N \ell$
2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$	$\eta_{a_2} \bar{\epsilon}_{a_1, a_2} \overline{y_{N, i_1, r_1}} N_{r_1}{}^T C P_L \ell_{a_1, i_1}$ $+ \epsilon_{a_2, a_1} \eta_{a_1}{}^\dagger y_{N, i_1, r_1} \overline{\ell_{a_2, i_1}} P_R N_{r_1}$
2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$	$\eta_{a_2} \bar{\epsilon}_{a_1, a_2} \overline{y_{N, i_1, r_1} \text{gen}(\text{SU}2\text{L}(\text{fund}))}_{A_1, a_1, a_3} N_{A_1, r_1}{}^T C P_L \ell_{a_3, i_1}$ $+ \epsilon_{a_2, a_1} \eta_{a_1}{}^\dagger y_{N, i_1, r_1} \text{gen}(\text{SU}2\text{L}(\text{fund}))_{A_1, a_3, a_2} \overline{\ell_{a_3, i_1}} P_R N_{A_1, r_1}$
3	$\frac{1}{2}$	2	0	$+D_{\mu_1}(\eta_{A_1}) D_{\mu_1}(\eta_{A_1})^\dagger$	$\eta_{A_1} \overline{y_{N, i_1, r_1} \text{gen}(\text{SU}2\text{L}(\text{fund}))}_{A_1, a_1, a_2} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1}$ $+ \eta_{A_1}{}^\dagger y_{N, i_1, r_1} \text{gen}(\text{SU}2\text{L}(\text{fund}))_{A_1, a_1, a_2} \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1}$